

Chapter 18

CS2 course package - Hawaii Pacific University (HPU)

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How to write your chapter using this Overleaf template

Welcome to the **Modern Course Exemplars infused with Parallel and Distributed Computing for CS1 and CS2 introductory computing courses** book project. To ensure your chapter integrates seamlessly into the final volume of the book, please follow these structure and formatting guidelines. Your contribution is managed as a modular component of the book; therefore, adhering to the standard template is essential for a successful compilation.

Important Note on Packages: If you find it absolutely necessary to add specific LaTeX packages or global command definitions, place all `\usepackage` commands and custom macros inside the `include.tex` file located outside of the chapter folders.

File Organization

You will find the following file structure:

- `cover.tex`: Your chapter title, authors, and affiliations.
- `content.tex`: Your abstract, PDC description, and main body text.
- `appendix.tex`: Supplementary materials, git link, and rubrics.
- `figures/`: All image files (`.png`, `.jpg`, `.pdf`).
- `references.bib`: Your BibTeX bibliography file.

Cover Page Configuration (**cover.tex**)

Each chapter begins with a title and author block. Please update the `cover.tex` file using the following template. Ensure the chapter title is updated in **both** the bracketed short title (used for the Table of Contents) and the main title.

Author List Formatting

To maintain consistency throughout the volume, please adhere to these naming rules:

- **Two Authors:** Use “and” between names (e.g., *John Smith¹ and Jane Doe²*).
- **Three or More Authors:** Use commas for all but the last author (e.g., *John Smith¹, Jane Doe², Bob White³*).
- **Affiliations:** Match the superscript numbers in the author list to the corresponding university in the lines below.

Figures and Tables

Mandatory Labeling Convention

To prevent cross-referencing conflicts when all chapters are merged, you **must** prefix every figure and table label with your assigned ***Chapter-ID***_, that is, your chapter ID followed by an underscore (for example, TTU2_, for this chapter by TTU in CS2).

- **Correct:** `\label{fig:TTU2_analysis}`
- **Incorrect:** `\label{fig:analysis}`

Tables

Authors should use the `table` environment with center-aligned columns and vertical lines. Every row must end with `\\` to avoid `noalign` errors.

Table 18.1: Potential speedup for search algorithms.

Algorithm	Sequential	Parallelizable
Linear Search	0%	100%

Figures

Place all images in the `figures/` sub-folder. Refer to them using the `figures/` prefix only; do **not** include the chapter folder name in the path. Use the `[htb]` float specifier. Images should typically be scaled relative to the text width (e.g., `0.8\linewidth`).

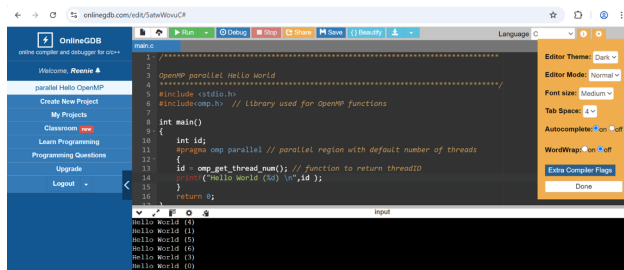


Figure 18.1: Description of the image content.

Citations

Use standard `\cite{label}` commands. Ensure all corresponding entries are in your `references.bib` file.

Abstract

|Your abstract text here.|

Descriptor Elements

Programming Language Employed: |e.g., Java, others C/C++|

Relevant core courses: |e.g., CS1, Data Structures|

Pre-requisites: |Required student background|

Relevant PDC topics: |List topics here with bloom classification (use K-knowledge, C-comprehension, or A-be able to apply)|

Relevant PDC topics and Blooms levels

PDC Concept	Chapter Section				
	1.1	1.2	1.3	1.4	1.5
Topic1	K	C	A		
Topic2	K	C	A		
Topic3		C	A	A	

Learning outcomes:

1. |Outcome 1|
2. |Outcome 2|

Context for use: |Your local context about institution, department, mix and levels of students, class size, language employed, etc.|

18.1 Introduction

|Motivation and background for the chapter.|

18.2 How CS1 was changed and PDC topics integrated

|Explain how the existing course (e.g., CS1) was modified.|

18.2.1 Integrated Syllabi (Pre and Post)

|Provide high level syllabi highlighting changes.|

18.2.2 Highlights

|What remained the same, and what specifically needed to be changed or deleted?|

18.2.3 Challenges

|Discuss logistical, pedagogical, or technical difficulties.|

18.3 New Unplugged Activities

18.3.1 Activity Name (e.g., Flag Maker)

Problem Description, Prerequisites, and Learning Outcomes

Implementation Details

- **Logistics** (Material required for the activity, etc.)
- **How to Conduct**
- **Instructional materiel** (Lectures, Videos, Assignments, etc.)

Experience and Teaching Tips

Data and Analysis

18.4 New Plugged-in Activities

18.4.1 Activity Name (e.g., Parallel Search)

Problem Description, Prerequisites, and Learning Outcomes

Algorithms

Implementation Details

- Instructional Material (Lectures, Videos, Assignments, Tests)
- How-To Guides

Experience and Teaching Tips

Data and Analysis

18.5 Course-wide Pre and Post Course Surveys

|Data collection and analysis of student feedback/learning.|

18.6 Other Activities and Ideas

|Alternative ways to apply these concepts, etc..|

18.7 Conclusion

Bibliography

Appendix

|You must include these four specific subsections (in addition to anything else):|

Github Link to Instructional Material

`https://github.com/...`

Detailed Pre and Post Syllabus

Organization of Material

|Explain the folder structure of your Github repository.|

Survey and Other Anonymized Data and Analysis

|Include student feedback, learning data, or implementation analysis.|